Curiosity in Hindsight: Intrinsic Exploration in Stochastic Environments

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Abstract

Consider the problem of exploration in sparse-reward or reward-free environments, such as in Montezuma's Revenge. In the *curiosity-driven* paradigm, the agent is rewarded for how much each realized outcome differs from their predicted outcome. But using predictive error as intrinsic motivation is fragile in *stochastic environments*, as the agent may become trapped by high-entropy areas of the state-action space, such as a "noisy TV". In this work, we study a natural solution derived from structural causal models of the world: Our key idea is to learn representations of the future that capture precisely the *unpredictable* aspects of each outcome—which we use as additional input for predictions, such that intrinsic rewards only reflect the *predictable* aspects of world dynamics. First, we propose incorporating such hindsight representations into models to disentangle "noise" from "novelty", yielding *Curiosity in Hindsight*: a simple and scalable generalization of curiosity that is robust to stochasticity. Second, we instantiate this framework for the recently introduced `BYOL-Explore` algorithm as our prime example, resulting in the noise-robust `BYOL-Hindsight`. Third, we illustrate its behavior under a variety of different stochasticities in a grid world, and find improvements over `BYOL-Explore` in hard-exploration Atari games with sticky actions. Notably, we show state-of-the-art results in exploring Montezuma's Revenge with sticky actions, while preserving performance in the non-sticky setting.

1. Introduction

Learning to understand the world without supervision is a hallmark of intelligent behavior [1], and *exploration* is a key pillar of research in reinforcement learning agents [2]. How might an agent learn meaningful behaviors when external rewards are sparse or absent? A predominant approach is given by the *curiosity-driven* paradigm [3], in which an agent's ability to predict the future is used as a proxy for their "understanding" of the world. Maintaining a learned model of the environment, at each step the agent receives an intrinsic reward proportional to how much the realized outcome differs from their predicted outcome—which naturally directs them towards new areas that have not been seen.

There are two major hurdles. The first concerns *dimensionality*: While outcomes can be predicted directly at the level of observations [4–8], pixel-based losses have generally not worked well in higher dimensions [9]. Popular solutions thus operate on lower-dimensional *latent representations*, such as frame-predictive features [10], inverse dynamics features [11], random features [12], or features that maximize information across time [13]. Most recently, bootstrapped features are employed in `BYOL-Explore` [14]—achieving superhuman performance on hard-exploration games in Atari with a much simpler design than comparable agents.

The second concerns *stochasticity*, which is our focus here: Curiosity-driven agents are often susceptible to bad behavior in environments with stochastic transitions, since they are often hopelessly attracted to high-entropy elements in the state-action space [9]. A classic example is the problem of a "noisy TV", which generates a stream of intrinsic rewards around which predictive error-based agents become stuck indefinitely [15]. More generally, this problem manifests with respect to any aspect of environment dynamics that is inherently unpredictable, including noise specific to certain states, as well as noise actively induced by the agent.

Novelty vs. Noise In the presence of stochasticity, predictive error *per se* is no longer a good measure for an agent's lack of "understanding" of the world. Intuitively, we wish to measure their understanding by how much *epistemic* knowledge they have acquired (viz. necessary truths about how the world works in general), which is entirely orthogonal to how much *aleatoric* variation each outcome can display (viz. contingent facts about how the world happens to be). Precisely, we want to distinguish between aspects of world dynamics that are inherently predictable—for which (reducible) errors stem from "novelty"—and aspects that are inherently unpredictable—for which (irreducible) errors stem from "noise". Crucially, while the former should contribute to intrinsic rewards for exploration, the latter should not.

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Contributions We operationalize this distinction by deriving a solution based on structural causal models of the world: Our key idea is to learn representations of the future that capture precisely the unpredictable aspects of each outcome—no more, no less—which we use as additional input for predictions, such that intrinsic rewards vanish in the limit. First, we propose incorporating such hindsight representations into the agent's model to disentangle "noise" from "novelty", yielding *Curiosity in Hindsight*: a simple and scalable generalization of curiosity-driven exploration that is robust to stochasticity (Section 3). Second, we instantiate this framework for the recently introduced `BYOL-Explore` algorithm as our prime example, giving rise to the noise-robust `BYOL-Hindsight` (Section 4). Third, we illustrate its behavior under a variety of different stochasticities in a grid world, and find improvements over `BYOL-Explore` in hard-exploration Atari games with sticky actions (a standard protocol for introducing stochasticity in training/evaluation). Notably, we show state-of-the-art results in exploring Montezuma's Revenge with sticky actions, while preserving its original performance in the non-sticky setting (Section 5).

2. Motivation

2.1. Problem Formalism

Consider the standard MDP setup. We employ uppercase for random variables and lowercase for specific values: Let X denote the *state* variable, taking on values $x \in \mathcal{X}$, and A the *action* variable, taking on values $a \in \mathcal{A}$. While we keep notation simple, X may play the role of "contexts", "features", "embeddings", or "beliefs" depending on environment observability and the design of the agent. Let $\tau \in \Delta(\mathcal{X})^{\mathcal{X}\times\mathcal{A}}$ denote the world's dynamics such that $X_{t+1} \sim \tau(\cdot|x_t, a_t)$, and $\pi \in \Delta(\mathcal{A})^{\mathcal{X}}$ the agent's policy such that $A_t \sim \pi(\cdot|x_t)$. Lastly, let ρ_π denote the distribution of states induced by π.

Definition 1 (Curiosity-driven Exploration) In this work we focus on *predictive error-based* curiosity—subsuming most popular approaches to curiosity. Intrinsic rewards are:

$$\mathcal{R}_\eta(x_t, a_t) \coloneqq -\mathbb{E}_{X_{t+1}\sim\tau(\cdot|x_t,a_t)} \log \tau_\eta(X_{t+1}|x_t, a_t) \quad (1)$$

where τ_η is the agent's world model[2] parameterized by η, and is trained using the trajectories collected by rolling out a policy that seeks to maximize this same prediction error:

$$\underset{\pi}{\overset{\text{(policy)}}{\text{maximize}}} \ \underset{\eta}{\overset{\text{(model)}}{\min}} \ \mathbb{E}_{\substack{X_t\sim\rho_\pi \\ A_t\sim\pi(\cdot|X_t)}} \mathcal{R}_\eta(X_t, A_t) \quad (2)$$

Example 1 (Bootstrapping Representations) As our key example, recall `BYOL-Explore` [14], a most recent and successful incarnation of this paradigm. The *prediction loss* for a given transition (x_t, a_t, x_{t+1}) is defined as the following:

$$\mathcal{L}_\eta^{\texttt{BYOL}}(x_t, a_t, x_{t+1}) \coloneqq \big\|x_{t+1} - \hat{x}_{t+1}\big\|_2^2 \quad (3)$$

and the (state-action) *prediction bonus* for the agent's policy:

$$\mathcal{R}_\eta^{\texttt{BYOL}}(x_t, a_t) \coloneqq \mathbb{E}_{X_{t+1}\sim\tau(\cdot|x_t,a_t)} \mathcal{L}_\eta^{\texttt{BYOL}}(x_t, a_t, X_{t+1}) \quad (4)$$

where the novelty of the method lies in the manner in which specific quantities are defined and learned: **(i.)** Input states are RNN "belief" representations $x_t \coloneqq b_t$ of previous actions $\{a_{t'}\}_{t'<t}$ and observation encodings $\{\omega(o_{t'})\}_{t'\leq t}$, where ω is an encoding function; **(ii.)** Target states are ℓ_2-normalized encodings $x_{t+1} \coloneqq \texttt{sg}(\omega_{\text{target}}(o_{t+1})/\|\omega_{\text{target}}(o_{t+1})\|_2)$ of future observations, with ω_{target} being an exponential moving average of ω, and `sg` denotes the stop-gradient operator; and **(iii.)** Predictions are ℓ_2-normalized transformations of current beliefs and actions: $\hat{x}_{t+1} \coloneqq h_\eta(b_t, a_t)/\|h_\eta(b_t, a_t)\|_2$, where h_η is a prediction function. Multi-step open-loop predictions are a straightforward extension. See Appendix C for a more detailed review of the `BYOL-Explore` algorithm.

Stochastic Traps In stochastic environments, Equation 1 does not converge to zero even with infinite experience: It converges to the entropy $\mathbb{H}[X_{t+1}|x_t, a_t]$, so the agent may become stuck on repeatedly experiencing (intrinsically rewarding) transitions where entropy is high. Instead, what we desire is a reward that converges to zero in the limit. The notion of "optimistic" exploration offers a hint of what might be possible—Consider constructing a reward that satisfies:

$$\mathcal{R}_\eta(x_t, a_t) \geq D_{\text{KL}}\big(\tau(X_{t+1}|x_t, a_t)\|\tau_\eta(X_{t+1}|x_t, a_t)\big) \quad (5)$$

which upper bounds the distance between the world and the agent's model. On the one hand, while Definition 1 verifies this, the bound fails to tighten even in the limit. On the other hand, it is hard to measure this distance directly, since the entropy term is by construction unknown. As it turns out, we shall later see that our proposed technique effectively gives a reward that verifies the inequality—and is tight in the limit.

2.2. Related Work

Our work inherits from the curiosity-driven paradigm [3–18], among which some methods have been designed with robustness to certain stochasticities in mind (Table 1). However, our method is uniquely characterized by the following:

1. **Stochasticity Types**: First, it is capable of handling all types of stochasticities in generality. Specifically, this includes stochasticity that is *entirely random* (e.g. a viewport polluted by noise sampled according to a distribution independent of states and actions), stochasticity that is *state-dependent* (e.g. a visible object that performs a random walk within the environment), as well as *action-dependent* (e.g. a layer of random pixels that only appears if sampled on demand by specific actions). For instance, previous works have found that inverse dynamics features can learn to filter out random noise [11], but may break down in the presence of action-dependent noise [9, 19].

[2] Note that this is only used for computing rewards, and need not be related to the underlying RL algorithm, which can be model-free.

Table 1. Relationship with Curiosity-driven Exploration. Curiosity in Hindsight is a drop-in modification on top of any prediction error-based method, and is characterized by being robust to different noises, being dynamics aware, and being general to any representation space.

Curiosity-driven Exploration Method	*Prediction Inputs*	*Prediction Target*	*Measure of Learning*	Random Noise	X-/A-Dep. Noise	Dynamics Awareness	**Representation Space**
AE [10]	X_t, A_t	X_{t+1}	$\mathcal{L}_\eta^{\text{predict}}$	✗	✗	✓	reconstructive
ICM [11]	X_t, A_t	X_{t+1}	$\mathcal{L}_\eta^{\text{predict}}$	✓	✗	✓	action predictive
EMI [13]	X_t, A_t	X_{t+1}	$\mathcal{L}_\eta^{\text{predict}}$	✗	✗	✓	MI-maximizing
RND [12]	X_{t+1}	$f_{\text{random}}(X_{t+1})$	$\mathcal{L}_\eta^{\text{predict}}$	✓	✓	✗	random projection
Dora [16]	X_t, A_t	const. zero	$\mathcal{L}_\eta^{\text{predict}}$	✓	✓	✗	pixel space
AMA [15]	X_t, A_t	X_{t+1}	$\mathcal{L}_\eta^{\text{predict}} - \text{Tr}(\hat{\Sigma}_{t+1})$	✓	✓	✓	pixel space
`BYOL-Explore` [14]	X_t, A_t	X_{t+1}	$\mathcal{L}_\eta^{\text{predict}}$	✗	✗	✓	bootstrapped
Curiosity in Hindsight (e.g. `BYOL-Hindsight`)	X_t, A_t, Z_{t+1}	X_{t+1}	$\mathcal{L}_{\theta,\eta}^{\text{reconstruct}} + \mathcal{L}_{\theta,\nu}^{\text{invariance}}$	✓	✓	✓	***any representation***

2. **Dynamics Awareness**: Second, it does not require entirely discarding dynamics learning. By way of contrast, consider purely frequency-oriented exploration strategies, such as learning to predict a random projection of observations [12], or simply to predict the constant zero [16]. As these are deterministic functions of their inputs, they are in principle resilient to stochasticity. But empirically they can still behave poorly in the presence of action-dependent stochasticities [15]: If the noise is sufficiently *diffuse*, the agent may never learn the function well, so in the absence of any other learning signal—such as the dynamics of the world—they may still become stuck [20].

3. **Generality and Scalability**: As a drop-in modification, it is generally be applicable to any underlying choice of representation space. In contrast, existing techniques capable of handling stochasticity are often tied to specific feature spaces, such as to employ inverse dynamics features [11], random features [12], or pixel-space features [15]—which may limit their flexibility of application. Moreover, unlike ensemble-based or disagreement-based techniques that require training a large number of models [19, 21, 22],[3] we shall see that incorporating hindsight is simpler and more *scalable* by only requiring the addition of an auxiliary component to the usual prediction loss.

Alternative paradigms for exploration have been proposed: Novelty-based methods encourage exploration on the basis of visitation counts [25], hashes [26], density estimates [27–30], and adversarial guidance [31, 32]; further extensions account for episodic memory [33–35] and the long-term value of exploratory actions [16, 36–38]. Knowledge-based methods encourage exploration on the basis of the agent's uncertainty about the world [19, 39], with most work focusing on estimating the information gain from different actions [21,22,40–46], or directly estimating learning progress [47–49]. Finally, diversity-based methods seek to maximize the state entropy [50–53], or to encourage learning diverse skills [54–65] and reaching different goals [66–70].

[3] Ensembles can in principle approximate uncertainty [19,21–23]; in practice, training and scaling is difficult with larger architectures, and models often converge prematurely to the same outputs [24].

3. Curiosity in Hindsight

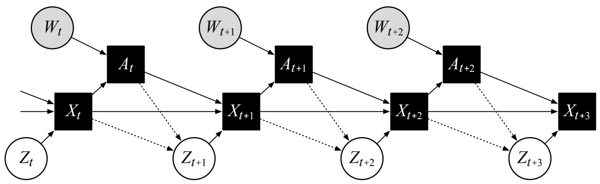
Consider the game of betting on a hidden dice roll: Suppose we take the action $A_t =$ "bet on 6", then observe the outcome $X_{t+1} =$ "lost the bet". Two facts are clear: (1) *a priori*, we could not have predicted this result at all; (2) *a posteriori*, we may deduce the (latent) fact $Z_{t+1} =$ "the die must have rolled 1–5". These are not contradictory. In particular, the former does *not* imply that we lack an understanding of how the game works, nor does it suggest that we should engage in further such bets to improve our understanding. Indeed, knowing how the game works, in hindsight (i.e. given what we deduced about Z_{t+1}), the outcome is obvious to us (i.e. we can now deterministically identify X_{t+1}). Conversely, suppose we actually *didn't* know how the game works: Then we couldn't have correctly inferred Z_{t+1}, nor would its knowledge have enabled us to identify X_{t+1} with any certainty. If so, engaging in additional bets may indeed allow us to learn and improve our understanding of how they work.

Intuitively, we can thus measure our understanding of each transition based on how much the outcome makes sense *in hindsight*. So, instead of asking "How well can we predict X_{t+1} *a priori*?", we actually want to ask "How well can we reconstruct X_{t+1} *a posteriori*—i.e. given hindsight Z_{t+1}?". First, we formalize this intuition using the language of *posterior inference* when a known model of the world is available (Section 3.1). Then we generalize this approach to generating learned *hindsight representations* when the world model needs to be learned at the same time (Section 3.2). Finally, we derive *Curiosity in Hindsight* on the basis of these ingredients, showing it approximates optimistic exploration (Inequality 5) while being robust to stochasticities (Section 3.3).

3.1. Structural Causal Model

Let Z denote a *latent* variable, taking on values $z \in \mathcal{Z}$. For each observed transition (x_t, a_t, x_{t+1}), we let z_{t+1} encapsulate *all* sources of unobserved stochasticity in the dynamics. By construction, $x_{t+1} = f(x_t, a_t, z_{t+1})$ for some deterministic function f, and a prior p over Z_{t+1} induces the environment dynamics—that is, the distribution $\tau(X_{t+1}|x_t, a_t)$. Figure 1 illustrates the structural causal model, with solid

Figure 1. Structural Causal Model. By the reparameterization lemma, there exists an equivalent graphical representation under which all stochasticities are exogenous (i.e. dotted edges removed).



squares for deterministic nodes, shaded circles for observable stochastic nodes, and unshaded circles for unobservable stochastic nodes (W captures any randomness in the policy). In general, stochasticities can be entirely random (i.e. no edges into Z_{t+1}), state-dependent (i.e. edge $X_t \to Z_{t+1}$), or action-dependent (i.e. edge $A_t \to Z_{t+1}$). However, by the *reparameterization lemma* it is always possible to represent an environment such that all stochasticities are effectively exogenous [71–73] (i.e. no directed edges into Z_{t+1}).

From Prediction to Reconstruction Consider the setting in which we know the model f. Suppose first that we somehow had access to each latent z_{t+1}. Then the outcome of a transition at state x_t and action a_t would be deterministically computable with no uncertainty (i.e. reconstruction error = zero):

$$x_{t+1} \equiv f(x_t, a_t, z_{t+1}) \tag{6}$$

In reality, the latent variable z_{t+1} is not observed. Thus it may seem like the best we can do is to compute the *a priori* expectation of the outcome (i.e. prediction error = entropy):

$$\mathbb{E}_{X_{t+1}\sim\tau(\cdot|x_t,a_t)} X_{t+1} \equiv \mathbb{E}_{Z_{t+1}\sim p} f(x_t, a_t, Z_{t+1}) \tag{7}$$

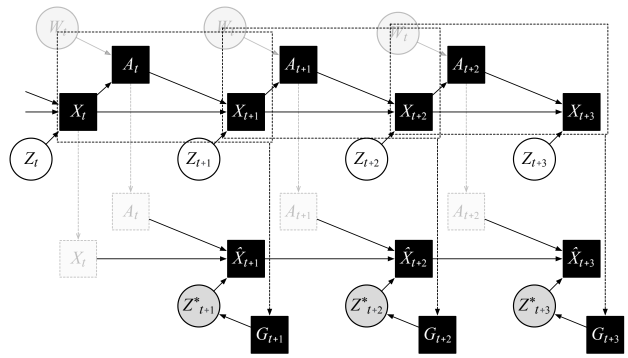
However, while z_{t+1} is not observable, based on the transition (x_t, a_t, x_{t+1}) we can infer *a posteriori* what its values could have been. Importantly, by the consistency property of counterfactuals we know $f(x_t, a_t, Z_{t+1}) = x_{t+1}$ for any $Z_{t+1} \sim p(\cdot|x_t, a_t, x_{t+1})$ [74]. That is to say, conditioned on hindsight, the reconstruction error of the true model is zero. This suggests when f is unknown and learned by the agent, *the reconstruction error may be an attractive candidate for an intrinsic reward*. Of course, now the missing piece is how to sample Z_{t+1} from the posterior—which we discuss next.

3.2. Hindsight Representations

Realistically, the model $f(X_t, A_t, Z_{t+1})$ is unknown, so we learn to approximate it using a *reconstructor* f_η parameterized by η.[4] Exact posterior inference $p_\eta(Z_{t+1}|X_t, A_t, X_{t+1})$ is intractable, so we learn to approximate it using a *generator* p_θ parameterized by θ. Two objectives are key. First, as noted above, representations Z_{t+1} should be *reconstructive* of outcomes X_{t+1}. Here we can simply use the squared loss:

[4]In general the model is not identifiable and there is no guarantee that $f_\eta(x_t, a_t, Z_{t+1})$ be close to x_{t+1} for arbitrary Z_{t+1}. But we are not interested in making counterfactual queries: For reconstruction, we only wish to evaluate f_η where $Z_{t+1} \sim p_\eta(\cdot|x_t, a_t, x_{t+1})$.

Figure 2. Hindsight Representations. A learned generator $G_{t+1} :=$ $p_\theta(\cdot|x_t, a_t, x_{t+1})$ generates hindsight vectors—denoted Z^*_{t+1} in this figure to be distinguished from "ground-truth" latents Z_{t+1}.



Objective 1 (Reconstruction) Let the *reconstruction loss* for a given transition $(x_t, a_t, z_{t+1}, x_{t+1})$—including hindsight representation z_{t+1} drawn from $p_\theta(\cdot|x_t, a_t, x_{t+1})$—be:

$$\mathcal{L}^{\text{rec.}}_\eta(x_t, a_t, z_{t+1}, x_{t+1}) := \left\| x_{t+1} - f_\eta(x_t, a_t, z_{t+1}) \right\|_2^2 \tag{8}$$

and (state-action) *reconstruction bonus* for the agent policy:

$$\mathcal{R}^{\text{rec.}}_{\theta,\eta}(x_t, a_t) := \mathbb{E}_{\substack{X_{t+1}\sim\tau(\cdot|x_t,a_t)\\ Z_{t+1}\sim p_\theta(\cdot|x_t,a_t,X_{t+1})}} \mathcal{L}^{\text{rec.}}_\eta(x_t, a_t, Z_{t+1}, X_{t+1}) \tag{9}$$

Driven to zero, this requires hindsight representations to encapsulate *at least* all aspects of the world's dynamics that are unpredictable (so that we *don't* reward the agent for irreducible error). However, we also don't want Z_{t+1} to simply leak information about the outcome that is actually predictable to begin with (so that we *do* reward the agent for reducible error).[5] Thus our second objective requires it to be *independent* of X_t, A_t. Denote the pointwise mutual information between state-action x_t, a_t and hindsight z_t by $\text{PMI}_\theta(x_t, a_t; z_{t+1}) := \log(p_\theta(z_{t+1}|x_t, a_t)/p_\theta(z_{t+1}))$. Then:

Objective 2 (Invariance) Let the *invariance loss* for a given transition $(x_t, a_t, z_{t+1}, x_{t+1})$—again, where the hindsight representation z_{t+1} is drawn from $p_\theta(\cdot|x_t, a_t, x_{t+1})$—be:

$$\mathcal{L}^{\text{inv.}}_\theta(x_t, a_t, z_{t+1}) := \text{PMI}_\theta(x_t, a_t; z_{t+1}) \tag{10}$$

and (state-action) *invariance bonus* for the agent's policy:

$$\mathcal{R}^{\text{inv.}}_\theta(x_t, a_t) := \mathbb{E}_{\substack{X_{t+1}\sim\tau(\cdot|x_t,a_t)\\ Z_{t+1}\sim p_\theta(\cdot|x_t,a_t,X_{t+1})}} \mathcal{L}^{\text{inv.}}_\theta(x_t, a_t, Z_{t+1}) \tag{11}$$

Driven to zero, this requires hindsight representations to encapsulate *at most* all aspects of the world's dynamics that are unpredictable. This suggests a combination—of reconstruction loss (of a dynamics model) plus invariance loss (of a hindsight model)—may make a good signal for exploration. We now have all the ingredients required for Curiosity in Hindsight, which it is instructive to contrast with Definition 1:

[5]For example, consider the solution $Z_{t+1} := X_{t+1}$, where reconstruction error is trivially zero, but is pathological for exploration.

3.3. Optimistic Exploration

Definition 2 (Curiosity in Hindsight) Let the *hindsight intrinsic reward function* be defined as the weighted combination of Objectives 1 and 2, with a tradeoff coefficient λ:

$$\mathcal{R}_{\theta,\eta}(x_t, a_t) := \frac{1}{\lambda}\mathcal{R}^{\text{rec.}}_{\theta,\eta}(x_t, a_t) + \mathcal{R}^{\text{inv.}}_{\theta}(x_t, a_t) \quad (12)$$

and the dynamics and hindsight models are jointly trained to minimize this quantity over the trajectories it collects, while rolling out a policy seeking to maximize this same quantity:

$$\underset{\pi}{\overset{\text{(policy)}}{\text{maximize}}} \; \underset{\theta,\eta}{\overset{\text{(model)}}{\min}} \; \mathbb{E}_{\substack{X_t \sim \rho_\pi \\ A_t \sim \pi(\cdot|X_t)}} \mathcal{R}_{\theta,\eta}(X_t, A_t) \quad (13)$$

Recall that in the presence of stochastic transitions, standard curiosity-driven exploration can be seen as a poor approximation to "optimistic" exploration (Inequality 5)—because the bound is never tight even in the limit, which renders it susceptible to stochastic traps. The following result shows that exploration by Curiosity in Hindsight can resolve this:

Theorem 1 (Optimistic Exploration) Let the coefficient λ satisfy the inequality $\frac{1}{2}\log(\lambda\pi) \leq \mathbb{H}_\theta[X_{t+1}|x_t, a_t, Z_{t+1}] + D_{\text{KL}}\big(p_\theta(Z_{t+1}|x_t, a_t)\|p_\theta(Z_{t+1})\big)$, where π denotes here the mathematical constant (not the agent's policy). Then:

$$\mathcal{R}_{\theta,\eta}(x_t, a_t) \geq D_{\text{KL}}\big(\tau(X_{t+1}|x_t, a_t)\|\tau_{\theta,\eta}(X_{t+1}|x_t, a_t)\big) \quad (14)$$

where $\tau_{\theta,\eta}(X_{t+1}|x_t, a_t) := \mathbb{E}_{Z_{t+1}\sim p_\theta} p_\eta(X_{t+1}|x_t, a_t, Z_{t+1})$ denotes the learned world model. Furthermore, assuming realizability, rewards vanish at optimal parameters θ^*, η^*:

$$\mathcal{R}_{\theta^*,\eta^*}(x_t, a_t) = 0 \quad \forall x_t, a_t \in \text{supp}(\rho_\pi) \quad (15)$$

Proof. Appendix A. □

In other words, by choosing a small enough λ term, the hindsight intrinsic reward (Equation 12) is an upper bound on the KL-term we care about (Inequality 5). Driving the reward to zero also drives the KL to zero, thus the reward-maximizing exploration policy (Equation 13) is approximating precisely the sort of "optimistic" exploration that we desired to start.

4. Practical Framework

Two questions remain. Firstly, how should the invariance terms be computed? For this, we propose a contrastive learning framework to approximate them (Section 4.1). Secondly, what does a concrete implementation look like? For this, we instantiate this framework on top of `BYOL-Explore`, yielding its robust variant `BYOL-Hindsight` (Section 4.2).

4.1. Contrastive Learning

To estimate the pointwise mutual information, we use an auxiliary *critic* g_ν parameterized by ν, trained as maximizer:

Objective 3 (Contrastive Learning) Let the *contrastive loss* for a transition, with respect to a batch of $K-1$ negative hindsight samples $z^{1:K-1}_{t+1} := z^1_{t+1}, ..., z^{K-1}_{t+1}$, be defined as:

$$\ell^{K,\text{con.}}_{\theta,\nu}(x_t, a_t, z_{t+1}, z^{1:K-1}_{t+1}) := \quad (16)$$
$$\log \frac{e^{g_\nu(x_t,a_t,z_{t+1})}}{\frac{1}{K}\left(e^{g_\nu(x_t,a_t,z_{t+1})} + \sum_{i=1}^{K-1} e^{g_\nu(x_t,a_t,Z^i_{t+1})}\right)}$$

such that the overall contrastive loss for the transition is its expectation over negative hindsight samples from rollouts:

$$\mathcal{L}^{K,\text{con.}}_{\theta,\nu}(x_t, a_t, z_{t+1}) :=$$
$$\mathbb{E}_{\substack{(X^1_t,...,X^{K-1}_t)\sim\prod_{i=1}^{K-1}\rho_\pi \\ (A^1_t,...,A^{K-1}_t)\sim\prod_{i=1}^{K-1}\pi(\cdot|X^i_t) \\ (X^1_{t+1},...,X^{K-1}_{t+1})\sim\prod_{i=1}^{K-1}\tau(\cdot|X^i_t,A^i_t) \\ (Z^1_{t+1},...,Z^{K-1}_{t+1})\sim\prod_{i=1}^{K-1}p_\theta(\cdot|X^i_t,A^i_t,X^i_{t+1})}} \ell^{K,\text{con.}}_{\theta,\nu}(x_t, a_t, z_{t+1}, z^{1:K-1}_{t+1}) \quad (17)$$

and (state-action) *contrastive bonus* for the agent's policy:

$$\mathcal{R}^{K,\text{con.}}_{\theta,\nu}(x_t, a_t) :=$$
$$\mathbb{E}_{\substack{X_{t+1}\sim\tau(\cdot|x_t,a_t) \\ Z_{t+1}\sim p_\theta(\cdot|x_t,a_t,X_{t+1})}} \mathcal{L}^{K,\text{con.}}_{\theta,\nu}(x_t, a_t, Z_{t+1}) \quad (18)$$

How does Objective 3 approximate Objective 2? Precisely:

Theorem 2 (Optimal Invariance) The contrastive bonus lower-bounds the (ideal) invariance bonus for any pair x_t, a_t:

$$\mathcal{R}^{K,\text{con.}}_{\theta,\nu}(x_t, a_t) \leq \mathcal{R}^{\text{inv.}}_{\theta}(x_t, a_t) \quad (19)$$

Furthermore, assuming realizability, for optimal critic parameter $\nu^*_K := \arg\max_\nu \mathbb{E}_{X_t,A_t\sim\rho_\pi}\mathcal{R}^{K,\text{con.}}_{\theta,\nu}(X_t, A_t)$ the bound is asymptotically tight (in the batch size $K \to \infty$):

$$\lim_{K\to\infty} \mathcal{R}^{K,\text{con.}}_{\theta,\nu^*_K}(x_t, a_t) = \mathcal{R}^{\text{inv.}}_{\theta}(x_t, a_t) \quad (20)$$

Proof. Appendix A. □

Practical Algorithm This suggests a straightforward algorithm. In practice, batch size $K < \infty$ and critic ν is not fully optimized. The intrinsic reward (Definition 2) now becomes:

$$\mathcal{R}^{K}_{\theta,\eta,\nu}(x_t, a_t) := \frac{1}{\lambda}\mathcal{R}^{\text{rec.}}_{\theta,\eta}(x_t, a_t) + \mathcal{R}^{K,\text{con.}}_{\theta,\nu}(x_t, a_t) \quad (21)$$

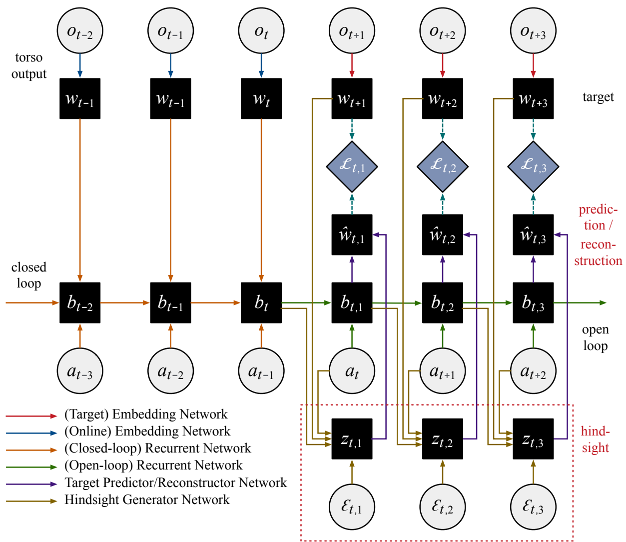
and optimization alternates between training the critic (maximizer) and dynamics and hindsight models (minimizers):

$$\underset{\pi}{\overset{\text{(policy)}}{\text{maximize}}} \; \underset{\theta,\eta}{\overset{\text{(model)}}{\min}} \; \underset{\nu}{\max} \; \mathbb{E}_{\substack{X_t \sim \rho_\pi \\ A_t \sim \pi(\cdot|X_t)}} \mathcal{R}^{K}_{\theta,\eta,\nu}(X_t, A_t) \quad (22)$$

Overall, our framework constitutes a simple drop-in modification on top of any curiosity-driven method: Instead of learning a *predictive model* specifying $X_{t+1} \sim \tau_\eta(\cdot|X_t, A_t)$, we now learn a (hindsight-augmented) *reconstructive model* specifying $X_{t+1} = f_\eta(X_t, A_t, Z_{t+1})$. The main ingredients include the reconstructor $f_\eta(X_t, A_t, Z_{t+1})$, the generator $p_\theta(Z_{t+1}|X_t, A_t, X_{t+1})$, and the critic $g_\nu(X_t, A_t, Z_{t+1})$, and the main hyperparameters are the contrastive batch size K and the coefficient λ in the hindsight intrinsic reward.

Finally, note that for ease of exposition we have focused on modeling single-step transitions; however, it is straightforward to generalize this approach to the case of multi-step reconstruction horizons using open-loop rollouts (Figure 2).

Figure 3. From BYOL-Explore to BYOL-Hindsight. The latter simply replaces the prediction loss with our reconstruction and contrastive losses (through the addition of hindsight), with no change to the underlying (bootstrapped) representation learning method.



4.2. BYOL-Hindsight

The preceding discussion used MDP notation, but in practice input states are often representations of histories, and observations are often represented in latent space. Recall `BYOL-Explore` (Example 1) is an incarnation of curiosity (Definition 1) that learns such representations. We now augment it with hindsight, giving rise to the novel `BYOL-Hindsight`:

Example 2 (Bootstrapping with Hindsight) Let the *hindsight loss* for a transition $(x_t, a_t, z_{t+1}, x_{t+1})$ be defined as:

$$\mathcal{L}^{K,\,\texttt{BYOL-Hind.}}_{\theta,\eta,\nu}(x_t, a_t, z_{t+1}, x_{t+1}) := \frac{1}{\lambda}\mathcal{L}^{\text{rec.}}_{\eta}(x_t, a_t, z_{t+1}, x_{t+1}) + \mathcal{L}^{K,\text{con.}}_{\theta,\nu}(x_t, a_t, z_{t+1}) \tag{23}$$

and the (state-action) *hindsight bonus* for the agent's policy:

$$\mathcal{R}^{K,\,\texttt{BYOL-Hind.}}_{\theta,\eta,\nu}(x_t, a_t) := \mathop{\mathbb{E}}_{\substack{X_{t+1}\sim\tau(\cdot|x_t,a_t)\\ Z_{t+1}\sim p_\theta(\cdot|x_t,a_t,X_{t+1})}} \mathcal{L}^{K,\,\texttt{BYOL-Hind.}}_{\theta,\eta,\nu}(x_t, a_t, Z_{t+1}, X_{t+1}) \tag{24}$$

where the key difference from `BYOL-Explore` lies in swapping out predictions for reconstructions. Specifically, input states x_t and target states x_{t+1} are defined just as before: **(i.)** Input states are RNN "belief" representations; and **(ii.)** Target states are ℓ_2-normalized encodings of future observations. However, instead of (target) predictions, we now have (target) reconstructions: **(iii.)** Reconstructions are ℓ_2-normalized transformations of beliefs, actions, and hindsight: $f_\eta(b_t, a_t, z_{t+1}) := h_\eta(b_t, a_t, z_{t+1})/\|h_\eta(b_t, a_t, z_{t+1})\|_2$, where h_η is a reconstruction function, and the contrastive loss encourages hindsight Z_{t+1} to be independent of B_t, A_t.

Figure 3 gives the concrete architecture for `BYOL-Explore`/`BYOL-Hindsight`, with the full multi-step horizon setup: First, an *online* embedding network ω encodes observations o_t into representations $w_t = \omega(o_t)$. A *closed-loop* RNN then computes representations b_t of histories up until each time step t. This is used to initialize an *open-loop* RNN that computes representations $b_{t,i}$ for horizon steps indexed as i. These representations are fed to a predictor/reconstructor network to output $\hat{w}_{t,i}$.[6] Finally, prediction/reconstruction targets are encoded with a *target* embedding network that is an exponential moving average of the online network. The prediction/reconstruction error $\mathcal{R}_{t,i}$ at each open-loop step is computed, and the intrinsic reward associated to each observed transition (o_s, a_s, o_{s+1}) is the sum of errors $\sum_{t+i=s+1}\mathcal{R}_{t,i}$. In `BYOL-Hindsight`, the generator samples $Z_{t,i}$ by taking noise $\varepsilon_{t,i}$ as input, and an additional critic (not pictured) encourages $Z_{t,i}$ to be independent of $B_{t,i-1}$ and A_{t+i-1}. See Algorithms 1–2 in Appendix C for details.

5. Experiments

Three questions deserve empirical study: **(a.) Effectiveness**: In stochastic environments, predictive error-based methods —e.g. `BYOL-Explore`—may fail. Does `BYOL-Hindsight` address the problem? **(b.) Robustness**: Is the method robust to the different types of stochasticities—i.e. independent noise, state-dependent noise, and action-dependent noise? **(c.) Non-specificity**: In environments with no stochasticity, hindsight should confer no benefit. Does `BYOL-Hindsight` manage to preserve the performance of `BYOL-Explore`?

Implementation In all experiments, we start from the same architecture/hyperparameters for `BYOL-Explore` as given in [14], including target network EMA, open-loop horizon, intrinsic reward normalization/prioritization, representation sharing, and VMPO [75] as the underlying RL algorithm. `BYOL-Hindsight` begins from the same setup. The generator, reconstructor, and critic networks are MLPs with three hidden layers of 512; the dimension of the generator noise ϵ and hindsight vector is 256; and $\lambda = 1$. Finally, where shown for reference, RND and ICM are also implemented exactly as described in [14]. See Appendix C for additional detail.

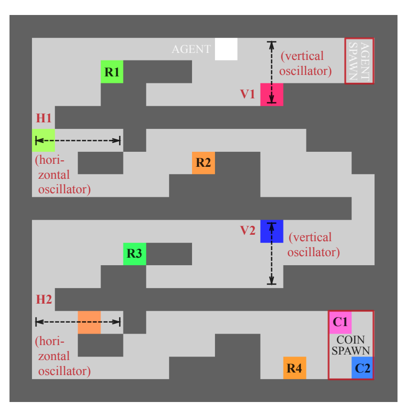


Figure 4. Pycolab Maze Environment Map.

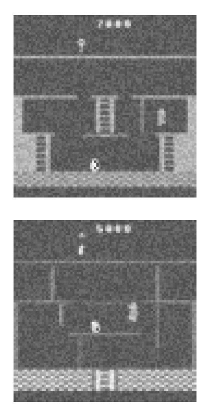
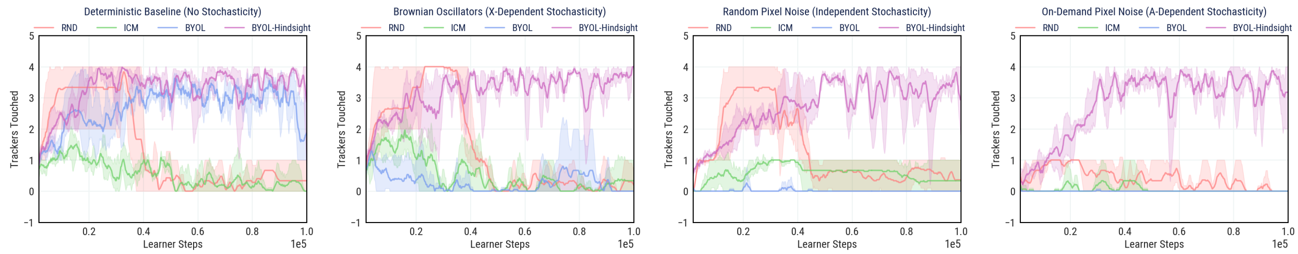


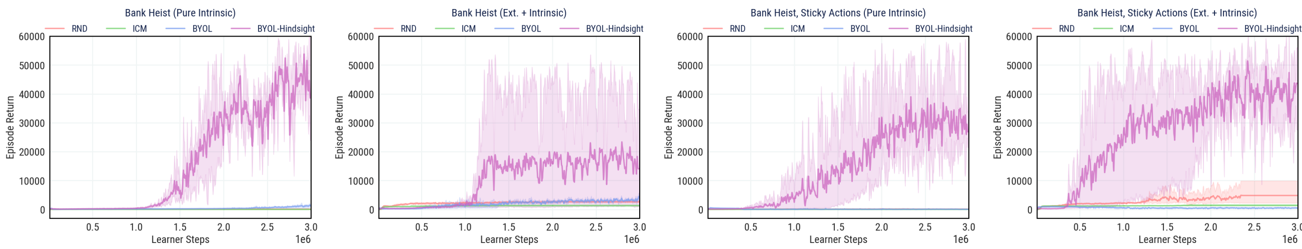
Figure 5. Pixel Noise.

[6]The composition of open-loop RNN and predictor/reconstructor networks is what we have abstractly denoted h_η in Examples 1–2.



(a) **Baseline: No Noise** (b) **Brownian Oscillators** (c) **Random Pixel Noise** (d) **On-Demand Pixel Noise**

Figure 6. Pycolab Maze, with Various Stochasticities. Performance measured by number of trackers touched in an episode (500-steps).



(a) **Natural Traps (Intrinsic Only)** (b) **Natural Traps (Ext. + Intrinsic)** (c) **Sticky Actions (Intrinsic Only)** (d) **Sticky Actions (Ext. + Intrinsic)**

Figure 7. Bank Heist, with Natural Traps and Sticky Actions. Performance measured by sum of extrinsic rewards obtained in an episode.

5.1. Pycolab Maze

First, to experiment with a variety of stochasticities in a controlled manner, we employ a Pycolab [76] maze (Figure 4): The agent spawns in the top right, and may explore past four (possibly stochastically oscillating) block elements (V1/2, H1/2), into the lower right where two coins are randomly spawned. The agent is purely intrinsically motivated, and progress is measured by trackers behind each of the block elements (R1–4). The agent only has access to a 5×5 frame (i.e. square radius 2) of its immediate surroundings as input.

Stochasticity We use four settings: "Baseline" (no noise); "Brownian Oscillators" (a form of state-dependent noise, where oscillators perform random walks along their axes of movements); "Random Pixel Noise" (a form of independent noise, which adds an extra layer of randomly sampled pixels to each frame with independent probability 0.25); as well as "On-Demand Pixel Noise" (a form of action-dependent noise, which does so whenever the no-op action is selected).

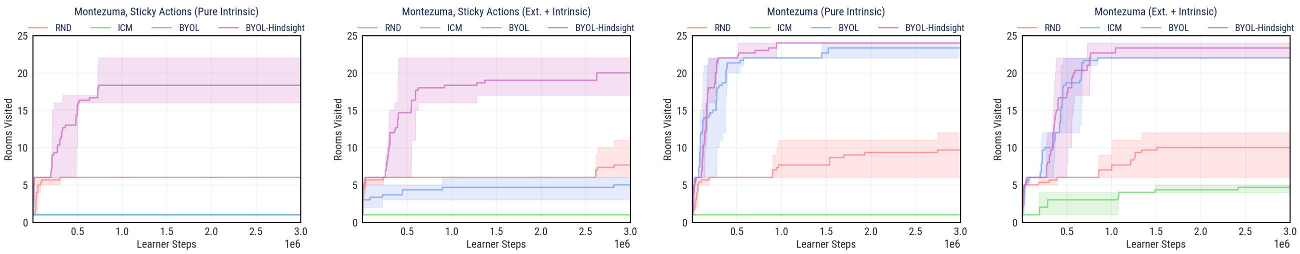
Results See Figure 6 (100k learner steps, 3 seeds). First, the "Baseline" setting tests non-specificity: Since there is no noise until the end, we expect curiosity-based exploration to do similarly with/without hindsight. For reference, we also show RND (in principle resilient to noise, as its targets are deterministic). All algorithms reach all four trackers (with RND eventually losing interest due to vanished rewards, as the environment is small). Second, in "Brownian Oscillators", `BYOL-Explore` fails to explore much beyond the first two trackers, as it is trapped by the unpredictable motion. In contrast, `BYOL-Hindsight` and RND both still explore the entire maze. Third, in "Random Pixel Noise" the results are similar, except both `BYOL-Explore` and RND do worse as the noise is an entire layer of random pixels (i.e. extremely diffuse), which outcompetes all other dynamics of the world in magnitude. Interestingly, while `BYOL-Hindsight` requires slightly longer to adapt, it still performs just as well. Lastly, the "On-Demand Pixel Noise" setting is most telling. `BYOL-Explore` is instantly trapped by the noise-inducing action, which it selects endlessly to generate intrinsic rewards. Even RND suffers greatly, which makes sense because the agent is no longer guaranteed a 0.75 probability of observing the world's unpolluted dynamics. In contrast, `BYOL-Hindsight` still performs as well as in the noise-free setting, underscoring robustness to different stochasticities.

5.2. Bank Heist

We use Atari with preprocessed grayscale 84×84-pixel images as input [77]. In all settings, we consider both "intrinsic-only" (no extrinsic signal) and "mixed" (extrinsic + intrinsic rewards) exploration regimes. In Bank Heist, the goal is to rob as many banks as possible while avoiding the police.

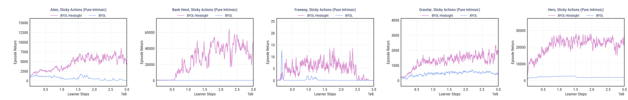
Stochasticity First, Bank Heist is characterized by naturally-occurring stochastic traps ("Natural Traps"), as noted by prior work [15]: It is impossible to predict where banks randomly regenerate and where bombs explode, thus a predictive error-based agent would simply endlessly enter and exit mazes while dropping bombs. Second, as an additional noise factor we use "Sticky Actions" [78] with stickiness 0.1.

Results See Figure 7 (3M learner steps, 3 seeds). Like in prior work, we measure the extrinsic reward per episode that the agent obtains, as a proxy for exploration ability. In both the "Natural Traps" and "Sticky Actions" settings, and in both intrinsic-only and mixed regimes, `BYOL-Explore`'s progress is immediately derailed by the stochastic traps, whereas `BYOL-Hindsight` achieves vastly better scores.

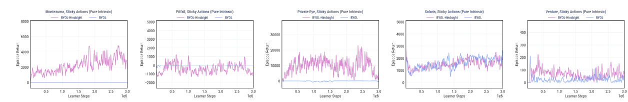


(a) **Sticky Actions (Intrinsic Only)** (b) **Sticky Actions (Ext. + Intrinsic)** (c) **Non-Sticky Baseline (Intrinsic Only)** (d) **Non-Sticky Baseline (Ext. + Intrinsic)**

Figure 8. Montezuma's Revenge, with Sticky Actions and Non-Sticky Baseline. Performance measured by rooms reached (episodic setting).



(a) **Alien (I)** (b) **Bank (I)** (c) **Freew. (I)** (d) **Gravitar (I)** (e) **Hero (I)**



(f) **Montez. (I)** (g) **Pitfall (I)** (h) **Priv. Eye (I)** (i) **Solaris (I)** (j) **Venture (I)**

Figure 9. Hard Exploration Games, with Sticky Actions. Performance measured by the sum of extrinsic rewards in an episode. Intrinsic-only "(I)" results shown; Figure 19 shows mixed "(E+I)".

5.3. Montezuma's Revenge

Playing Montezuma's Revenge requires learning complex dynamics including navigating around timed traps and moving enemies, and collecting keys to open doors in sequence.

Stochasticity First, Montezuma's Revenge is largely deterministic, which forms a natural baseline for testing non-specificity in a challenging exploration setting. Second, we use "Sticky Actions" similar to above to add stochasticity.

Results See Figure 8 (3M learner steps, 3 seeds). Like in prior work, exploration is measured by the number of different dungeon rooms the agent manages to discover over its lifetime. In "Sticky Actions", `BYOL-Explore` instantly flatlines in the intrinsic-only regime, and only does marginally better in the mixed regime. In contrast, `BYOL-Hindsight` explores most of the rooms in both regimes—which is an unprecedented result. In the "Non-Sticky Baseline", in both regimes `BYOL-Hindsight` does as well as the original performance of `BYOL-Explore`, which verifies non-specificity. See Appendix B.1 for evaluation results on episode return.

5.4. Hard Exploration Games

We conduct broad-based experiments for the ten hardest exploration games in Atari, where stochasticity is introduced with sticky actions as above. See Figure 9 for results in the intrinsic-only "(I)" regime, and also Figure 19 for the mixed "(E+I)" regime (3M learner steps, 1 seed). Like in prior work, we use extrinsic reward as a proxy for "interesting behavior". In the large majority of cases, `BYOL-Hindsight` improves over `BYOL-Explore`, especially when the latter simply flatlines. See Appendix B.7 for additional results and analysis.

5.5. Persistive Noise

While sticky actions is the standard protocol for adding stochasticity, we further design an especially challenging setting by corrupting observations with an additive layer of 84×84 pixel noise that persists across time as an action-triggerable random walk (Figure 5). See Appendix B.2 for full details and results, again verifying the robustness of the algorithm.

5.6. Temperature Sensitivity

The inner term in the contrastive loss (Objective 3) shares a similar form with contrastive losses in unsupervised representation learning [79–82], which admits a temperature parameter controlling the strength of penalties on negative samples [83]. See Appendix B.3 for a sensitivity analysis of the temperature parameter on the effectiveness of the algorithm.

5.7. Analysis of Invariance

For insight into invariance properties of the learned hindsight representations, see Appendix B.4 for analysis of intrinsic rewards over training; see Appendix B.5 for a comparison of prediction loss, reconstruction loss, and hindsight-only loss; and see Appendix B.6 for visualizations of what information the hindsight representations may be encoding.

6. Conclusion

In this work, we studied the problem that stochasticity poses to predictive error-based exploration. Theoretically, we refined our notion of curiosity to separate (learnable) epistemic knowledge from (unlearnable) aleatoric variation. Algorithmically, we proposed a method to learn (future-summarizing) representations of hindsight disentangled from (history-summarizing) representations of context. Practically, we arrived at a simple and scalable framework for generating (reducible) intrinsic rewards even in the presence of (irreducible) stochastic traps—without estimating the problematic entropy term at all. Our perspective shares connections with counterfactuals in policy evaluation [71–73], credit assignment [84–86], invariance [87–89], and fairness [90–93]. Future work may study explicitly generative world models to map stochastic latents to outcomes, as well as assessing the benefit of the approach in other stochastic domains such as NetHack. See Appendix D for an extended discussion.

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